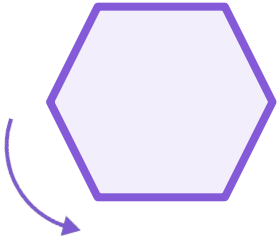




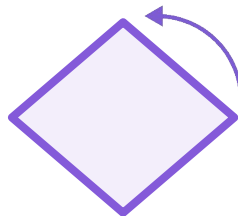
Describing a rotation's direction and size

Task A: Describing the direction of a rotation

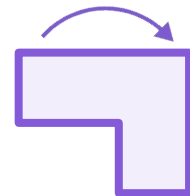
1) In which direction does this arrow suggest the rotation is happening?



2) In which direction does this arrow suggest the rotation is happening?

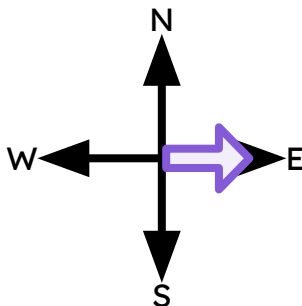


3) In which direction does this arrow suggest the rotation is happening?



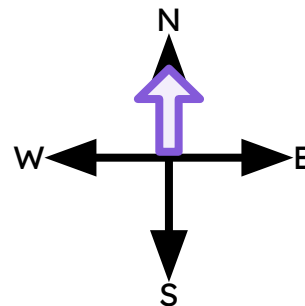
4) An arrow is pointing E (east).

If the arrow rotates anti-clockwise, which compass direction will it pass first?



5) An arrow is pointing N (north).

The arrow will rotate clockwise until it points N (north) again. In what order will it pass through the other compass directions?



Task B: Describing the size of a rotation

1) Write down the direction (up, down, left, right) each arrow will be pointing in after the rotation has been applied.

a) 90° rotation
anti-clockwise



b) 180° rotation
clockwise



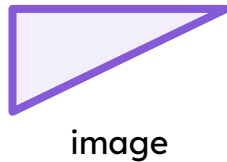
c) 270° rotation
anti-clockwise



d) 90° rotation
clockwise



2) The object has been rotated by less than 360° to create its image.
Complete the two sentences that could describe this transformation.



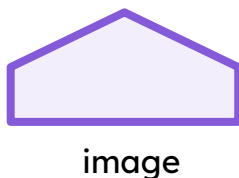
Possible Statement 1:

The object has been rotated _____ clockwise.

Possible Statement 2:

The object has been rotated 270° _____.

3) The object has been rotated by less than 360° to create its image.
Complete the two sentences that could describe this transformation.



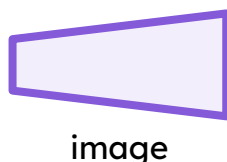
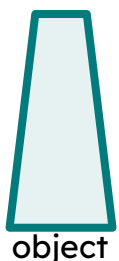
Possible Statement 1:

The object has been rotated 90° _____.

Possible Statement 2:

The object has been rotated _____.

4) The object has been rotated by less than 360° to create its image.
Complete the two sentences that could describe this transformation.



Possible Statement 1:

The object has been rotated _____ clockwise.

Possible Statement 2:

The object has been _____
anti-clockwise.

Task C: Describing more complex sizes of rotation

1) Write down the calculation and single **rotation** angle in degrees to each of the following statements.

a) between a quarter turn and a half turn

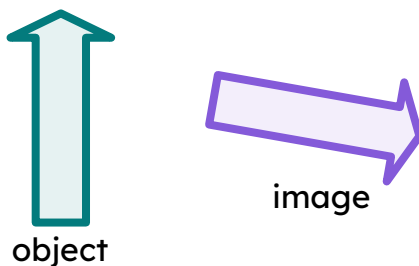
c) between no **rotation** and a quarter turn

b) two full turns

d) one and three-quarter turns

2) The object has been rotated by less than 360° to create its image.

Complete the two sentences that could describe this transformation.



Possible Statement 1:

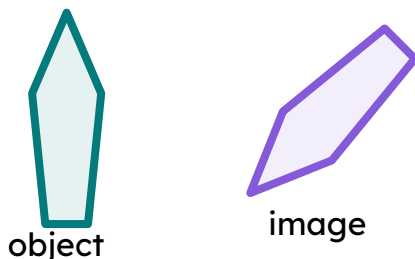
The object has been rotated 100° clockwise.

Possible Statement 2:

The object has been rotated _____ anti-clockwise.

3) The object has been rotated by less than 360° to create its image.

Complete the two sentences that could describe this transformation.



Possible Statement 1:

The object has been rotated 140° _____.

Possible Statement 2:

The object has been rotated _____.

4) This arrow will be rotated anti-clockwise through two full turns and three-quarters of another.



a) Which direction will the image of the arrow be pointing?

c) Give a description of a different anti-clockwise rotation that would result in the arrow pointing in the same direction.

b) In total, how many degrees will the arrow have rotated?

d) Give a description of a clockwise rotation that would result in the arrow pointing in the same direction.

